The Ben–Sasson–Sudan Reed–Solomon PCPP

A short, self-contained proof

August 2026

Abstract

We give a self-contained proof of the fixed-rate Reed–Solomon PCP of proximity used by Ben–Sasson and Sudan. A degree-$< N/8$ polynomial on an additive subspace is represented by a bivariate polynomial whose two individual degrees are about $\sqrt{N}$. Its evaluations form a word in a two-factor tensor product of Reed–Solomon codes. The prover supplies enough additional evaluations to connect the original word to this tensor-product word, and the verifier recursively tests one random coordinate slice. Soundness rests on the Polishchuk–Spielman theorem: a table whose two families of coordinate slices are close to Reed–Solomon codewords is close to one bivariate polynomial. We prove this theorem using only interpolation, greatest common divisors, a determinant, rank, and univariate root counting. The resulting proof has length $N(\log N)^{O(1)}$, uses $\log N + O(\log\log N)$ random bits and a constant number of field queries, has perfect completeness, and has inverse-polylogarithmic distance soundness.

1 Overview

Let L be an affine $\mathbb{F}_2$-subspace of a field $\mathbb{F} = \mathbb{F}_{2^\ell}$, and let $N = |L|$. For a finite set $S \subseteq \mathbb{F}$, write

$$\mathrm{RS}_{1/8}(S) = \{P|_S : \deg P < |S|/8\}.$$

All distances below are relative Hamming distances.

Theorem 1.1 (Reed–Solomon PCPP)**.** *There is an absolute constant C with the following property. Let $L \subseteq \mathbb{F}_{2^\ell}$ be an affine $\mathbb{F}_2$-subspace of dimension k, and assume $\ell \geq k+3$. There is a nonadaptive verifier with oracle access to a word $f : L \to \mathbb{F}$ and an auxiliary proof such that*

$$\Pr[\text{reject}] \geq \frac{\Delta_L(f, \mathrm{RS}_{1/8}(L))}{(k+1)^C}.$$

If $f \in \mathrm{RS}_{1/8}(L)$, some proof is accepted with probability one. The auxiliary proof has at most

$$|L|(k+1)^C$$

field symbols, the verifier uses $k + O(\log k)$ random bits, and it reads only a constant number of field symbols.

1.1 Why a tensor product appears

After translating the evaluation set, suppose that L is linear. Choose an $\mathbb{F}_2$-basis of the ambient field adapted to L, and split the coordinates of L into two blocks:

$$L = U \oplus V, \qquad |U| \asymp |V| \asymp \sqrt{N}.$$

The polynomial

$$q_U(T) = \prod_{u \in U} (T - u)$$

is additive, has kernel U, and is therefore constant on each coset $U + v$. It labels the cosets of U inside L.

Every polynomial P of degree less than $N/8$ has a unique representation

$$P(T) = Q(T, q_U(T)), \qquad \deg_X Q < |U|, \qquad \deg_Y Q < |V|/8. \tag{1.1}$$

On a coset $U + v$, the second argument of Q is fixed:

$$P(x) = Q(x, q_U(v)) \qquad (x \in U + v). \tag{1.2}$$

Thus one univariate polynomial of degree $\Theta(N)$ becomes a bivariate polynomial with individual degrees $O(\sqrt{N})$. Evaluating Q on a product set gives a word in a tensor product

$$C_A \otimes C_B,$$

where C_A and C_B are smaller Reed–Solomon codes. The verifier recursively checks one random A-slice or one random B-slice. A recursive step therefore reduces the problem size from N to $O(\sqrt{N})$.

The only serious question is local-to-global consistency: if the two families of slices are individually close to Reed–Solomon codewords, must they be close to one word of $C_A \otimes C_B$? For two Reed–Solomon codes the answer is yes, with a constant independent of the side lengths. This is the Polishchuk–Spielman theorem.

1.2 Why the two-factor theorem is special

Repeated tensor powers of three or more factors admit general robust tests: one tests a random codimension-one coordinate plane and uses the remaining coordinate directions to propagate local disagreement; see Viderman [4]. The analogous row–column statement is false for arbitrary pairs of good linear codes [5]. The two-factor Reed–Solomon case is therefore not an instance of a generic tensor-code theorem. Its algebraic structure supplies the missing local-to-global principle.

This distinction is also quantitative. The Ben–Sasson–Sudan recursion needs a constant robustness loss when the two side lengths are about $\sqrt{N}$. The special Reed–Solomon theorem provides exactly that. Once it is available, the rest of the construction is an elementary recursion.

2 Additive coordinates and the bivariate representation

2.1 An adapted basis

Write the original evaluation set as $a_0 + L_0$. Replacing $f(x)$ by $f(a_0 + x)$, and replacing a polynomial $P(T)$ by $P(a_0 + T)$, preserves ordinary polynomial degree. We may therefore assume that the evaluation set is a linear subspace L.

Regard $\mathbb{F}$ only as an ℓ-dimensional vector space over $\mathbb{F}_2$, and choose a basis

$$e_1, \dots, e_\ell$$

such that

$$L = \operatorname{span}_{\mathbb{F}_2}\{e_1, \dots, e_k\}. \tag{2.1}$$

This is a choice of notation for the additive subspaces; it is not a change of variable in the polynomial. In particular, we do not use the false assertion that ordinary univariate degree is invariant under an arbitrary $\mathbb{F}_2$-linear automorphism of $\mathbb{F}$.

Put

$$r = \lfloor k/2 \rfloor, \qquad s = \lceil k/2 \rceil,$$

and define the two coordinate blocks

$$U = \operatorname{span}_{\mathbb{F}_2}\{e_1, \dots, e_r\}, \qquad V = \operatorname{span}_{\mathbb{F}_2}\{e_{r+1}, \dots, e_k\}. \tag{2.2}$$

Then $L = U \oplus V$. At every recursive call we make the same normalization for the current affine evaluation space.

2.2 Subspace polynomials

For an $\mathbb{F}_2$-linear subspace $U \leq \mathbb{F}$, define

$$q_U(T) = \prod_{u \in U} (T - u).$$

Lemma 2.1 (Subspace polynomial)**.** *The polynomial q_U is monic of degree $|U|$, and*

$$q_U(x + y) = q_U(x) + q_U(y) \qquad (x, y \in \mathbb{F}).$$

Consequently $q_U : \mathbb{F} \to \mathbb{F}$ is $\mathbb{F}_2$-linear and has kernel U.

Proof. The degree, monicity, and root set are immediate. For additivity, induct on $\dim U$. If

$$U = U' \oplus \operatorname{span}_{\mathbb{F}_2}\{u_0\},$$

then, using the induction hypothesis,

$$q_U(T) = q_{U'}(T) q_{U'}(T + u_0) = q_{U'}(T)^2 + q_{U'}(u_0) q_{U'}(T).$$

The last expression is additive in characteristic two. □

Since $L = U \oplus V$ and $\ker q_U = U$, the restriction of q_U to V is injective. Hence

$$B := q_U(V) \tag{2.3}$$

is a linear subspace with $|B| = |V|$, and

$$q_U(u + v) = q_U(v) \qquad (u \in U,\ v \in V). \tag{2.4}$$

Thus q_U is the quotient coordinate for the decomposition $L = U \oplus V$.

2.3 Reducing the individual degrees

Lemma 2.2 (Bivariate representation)**.** *Let $q(T)$ be a monic polynomial of degree m. Every polynomial $P(T)$ of degree less than md can be written uniquely as*

$$P(T) = Q(T, q(T)),$$

where

$$\deg_X Q < m, \qquad \deg_Y Q < d.$$

Proof. The md polynomials

$$T^i q(T)^j, \qquad 0 \le i < m, \quad 0 \le j < d,$$

have distinct leading degrees $i + mj$, namely $0, 1, \ldots, md - 1$. They therefore form a basis for the polynomials of degree less than md. □

Let

$$m = |U|, \qquad n = |V|, \qquad N = mn.$$

Taking $q = q_U$ and $d = n/8$, every polynomial P of degree less than $N/8$ has a unique representation

$$P(T) = Q(T, q_U(T)), \qquad \deg_X Q < m, \qquad \deg_Y Q < n/8. \tag{2.5}$$

For $x = u + v \in U + v$, equations (2.4) and (2.5) give

$$P(x) = Q(x, q_U(v)). \tag{2.6}$$

This is the algebraic identity used throughout the recursion.

3 The recursive proof

Keep the notation above. Since $\ell \ge k+3$, the adapted basis contains three elements outside L. Set

$$w_1 = e_{k+1}, \qquad w_2 = e_{k+2}, \qquad w_3 = e_{k+3}.$$

Define the common first-coordinate set

$$A = w_3 + U + \mathrm{span}_{\mathbb{F}_2}\{w_1, w_2\}. \tag{3.1}$$

It is an affine subspace of size $4m$, disjoint from L.

For every $v \in V$, define an affine subspace

$$R_v = v + \big(U + \mathrm{span}_{\mathbb{F}_2}\{w_1, w_2, w_3 - v\}\big). \tag{3.2}$$

It has size $8m$ and contains both $U + v$ and A. The first inclusion is immediate. If

$$x = w_3 + u + \alpha w_1 + \beta w_2 \in A,$$

then

$$x = v + \big(u + \alpha w_1 + \beta w_2 + (w_3 - v)\big) \in R_v.$$

The product set on which the two tensor directions meet is

$$A \times B. \tag{3.3}$$

The original word is embedded into the set

$$\Gamma = \{(x, q_U(x)) : x \in L\} = \bigcup_{v \in V} (U + v) \times \{q_U(v)\}. \tag{3.4}$$

The map $x \mapsto (x, q_U(x))$ is a bijection from L to Γ.

Let

$$\mathcal{D} = \bigcup_{v \in V} R_v \times \{q_U(v)\}. \tag{3.5}$$

The first part of the auxiliary proof is a function

$$h : \mathcal{D} \setminus \Gamma \to \mathbb{F}.$$

Together with the input word, it defines a function $\widehat{h} : \mathcal{D} \to \mathbb{F}$: on Γ use the corresponding value of f, and elsewhere use h.

The rest of the proof contains recursive proximity proofs for the following words:

$$\widehat{h}(\cdot, q_U(v)) : R_v \to \mathbb{F} \qquad (v \in V), \tag{3.6}$$

and

$$\widehat{h}(x, \cdot) : B \to \mathbb{F} \qquad (x \in A). \tag{3.7}$$

The first family is tested against $\mathrm{RS}_{1/8}(R_v)$; the second is tested against $\mathrm{RS}_{1/8}(B)$.

3.1 Verifier

Fix a constant $k_0 \geq 8$. On an evaluation space of dimension at most k_0, the verifier reads the entire current word and checks Reed–Solomon membership directly. Otherwise it flips one fair coin.

If the outcome selects the first tensor direction, it chooses $v \in V$ uniformly and recursively verifies the word in (3.6). If the outcome selects the second tensor direction, it chooses $x \in A$ uniformly and recursively verifies the word in (3.7). Only one recursive call is made.

3.2 Honest proof and perfect completeness

Suppose

$$f = P|_L, \qquad \deg P < N/8.$$

Write $P(T) = Q(T, q_U(T))$ as in Theorem 2.2. The honest prover sets

$$h(x, y) = Q(x, y) \qquad ((x, y) \in \mathcal{D} \setminus \Gamma)$$

and recursively supplies honest proofs for all words in (3.6) and (3.7).

For fixed v, the polynomial

$$x \longmapsto Q(x, q_U(v))$$

has degree less than $m = |R_v|/8$. For fixed $x \in A$, the polynomial

$$y \longmapsto Q(x, y)$$

has degree less than $n/8 = |B|/8$. On Γ, equation (2.6) shows that these evaluations agree with f. Induction gives acceptance with probability one.

We have now described the proof system completely. Soundness requires one theorem about the tensor product of the two Reed–Solomon codes appearing on $A \times B$.

4 Robust testing of a two-factor Reed–Solomon tensor code

Let $A, B \subseteq \mathbb{F}$ be finite sets. Let

$$C_A = \{p|_A : \deg p \le d_A\}, \qquad C_B = \{q|_B : \deg q \le d_B\}.$$

Their tensor product $C_A \otimes C_B \subseteq \mathbb{F}^{A \times B}$ consists of the evaluations of bivariate polynomials $Q(X, Y)$ with

$$\deg_X Q \le d_A, \qquad \deg_Y Q \le d_B.$$

Equivalently, every slice $F(\cdot, b)$ belongs to C_A, and every slice $F(a, \cdot)$ belongs to C_B.

For an arbitrary table $F : A \times B \to \mathbb{F}$, define

$$\delta_A(F) = \mathbb{E}_{b \in B} \Delta_A(F(\cdot, b), C_A), \qquad \delta_B(F) = \mathbb{E}_{a \in A} \Delta_B(F(a, \cdot), C_B).$$

Theorem 4.1 (Polishchuk–Spielman)**.** *Suppose*

$$\frac{1}{4} \le \frac{|A|}{|B|} \le 4, \qquad d_A < |A|/4, \qquad d_B < |B|/8. \tag{4.1}$$

There is an absolute constant C_0 such that every table $F : A \times B \to \mathbb{F}$ satisfies

$$\Delta_{A \times B}(F, C_A \otimes C_B) \le C_0\big(\delta_A(F) + \delta_B(F)\big). \tag{4.2}$$

One may take $C_0 = 1024$.

Choose, for every $b \in B$, a nearest polynomial $r_b \in C_A$ to the slice $F(\cdot, b)$, and, for every $a \in A$, a nearest polynomial $c_a \in C_B$ to $F(a, \cdot)$. The two local descriptions may disagree. Define

$$D = \{(a, b) : r_b(a) \ne c_a(b)\}. \tag{4.3}$$

Every point of D is already an error of at least one local approximation, so

$$|D| \le \big(\delta_A(F) + \delta_B(F)\big)|A||B|. \tag{4.4}$$

Thus Theorem 4.1 reduces to a gluing statement: a sparse disagreement set between low-degree slices must have one global bivariate explanation.

4.1 The gluing statement

Theorem 4.2 (Gluing low-degree slices)**.** *For each $b \in B$, let $r_b(X)$ have degree at most d_A, and for each $a \in A$, let $c_a(Y)$ have degree at most d_B. Let*

$$D = \{(a, b) : r_b(a) \neq c_a(b)\}, \qquad s = |D|.$$

Suppose there are integers $e_A, e_B \geq 0$ such that

$$(e_A + 1)(e_B + 1) > s \tag{4.5}$$

and

$$\frac{e_A + d_A}{|A|} + \frac{e_B + d_B}{|B|} < 1. \tag{4.6}$$

Then there is a polynomial $Q(X, Y)$ with

$$\deg_X Q \leq d_A, \qquad \deg_Y Q \leq d_B,$$

such that, at all but at most $s + e_A e_B$ points of $A \times B$, both local polynomials give the value $Q(a, b)$.

The proof has four steps. First choose a nonzero low-bidegree polynomial $E(X, Y)$ that vanishes on D. Multiplying every local polynomial by E makes the two local descriptions agree at every grid point. Interpolation then gives one bivariate polynomial H realizing these products on all slices. Finally, fiberwise divisibility forces $E \mid H$; after dividing by E, only the few slices on which E vanishes identically can remain exceptional.

The first and third steps require the following two elementary lemmas.

4.2 Exact interpolation on a product set

Lemma 4.3 (Product interpolation)**.** *Let $G : A \times B \to \mathbb{F}$. Suppose every slice $G(\cdot, b)$ is represented on A by a polynomial of degree at most $D_A < |A|$, and every slice $G(a, \cdot)$ is represented on B by a polynomial of degree at most $D_B < |B|$. Then there is a unique polynomial $H(X, Y)$ with*

$$\deg_X H \leq D_A, \qquad \deg_Y H \leq D_B,$$

whose evaluations on $A \times B$ equal G. Its restrictions to the individual slices are the prescribed univariate polynomials.

Proof. Choose $D_A + 1$ points $a_0, \ldots, a_{D_A} \in A$, with Lagrange basis $L_0(X), \ldots, L_{D_A}(X)$. Let $g_i(Y)$ be the degree-$\leq D_B$ polynomial representing the slice through a_i. Set

$$H(X, Y) = \sum_{i=0}^{D_A} L_i(X) g_i(Y).$$

For fixed b, this polynomial agrees with the prescribed first-coordinate slice at $D_A + 1$ points, and both have degree at most D_A; hence they are equal. The second-coordinate statement and uniqueness follow in the same way. □

4.3 Fiberwise divisibility

Lemma 4.4 (Rank deficiency gives root multiplicity)**.** *Let $M(T) \in \mathbb{F}[T]^{N\times N}$. If*

$$\operatorname{rank} M(t) \le N - r,$$

then

$$(T-t)^r \mid \det M(T).$$

Proof. Put $S = T - t$. By constant invertible row and column operations, arrange

$$M(t) = \begin{pmatrix} I_j & 0 \\ 0 & 0 \end{pmatrix}, \qquad j \le N - r.$$

In $M(t+S)$, every entry outside the upper-left $j \times j$ block is divisible by S. A determinant monomial uses at most j entries from that block, and therefore at least $N - j \ge r$ entries divisible by S. □

Lemma 4.5 (Fiberwise divisibility)**.** *Let $E, H \in \mathbb{F}[X,Y]$, with $E \neq 0$. Let*

$$a_X, a_Y, b_X, b_Y, d_X, d_Y$$

be nonnegative integers satisfying

$$d_X = b_X - a_X, \qquad d_Y = b_Y - a_Y, \tag{4.7}$$

and suppose

$$\deg_X E \le a_X, \quad \deg_X H \le b_X, \qquad \deg_Y E \le a_Y, \quad \deg_Y H \le b_Y. \tag{4.8}$$

Assume that for every $b \in B$ there is a polynomial $r_b(X)$ of degree at most d_X such that

$$H(X,b) = E(X,b) r_b(X), \tag{4.9}$$

and for every $a \in A$ there is a polynomial $c_a(Y)$ of degree at most d_Y such that

$$H(a,Y) = E(a,Y) c_a(Y). \tag{4.10}$$

If

$$\frac{b_X}{|A|} + \frac{b_Y}{|B|} < 1, \tag{4.11}$$

then $E \mid H$ in $\mathbb{F}[X,Y]$. The quotient has X-degree at most d_X and Y-degree at most d_Y.

Proof. We first tighten the degree bounds without changing d_X or d_Y. If both $\deg_X E < a_X$ and $\deg_X H < b_X$, decrease a_X and b_X by one; repeat until at least one of the two degree bounds is attained. This preserves (4.7), (4.8), (4.9), and strictly improves (4.11). Do the same in the Y-variable. We may therefore assume that

$$\deg_X E = a_X \text{ or } \deg_X H = b_X, \qquad \deg_Y E = a_Y \text{ or } \deg_Y H = b_Y. \tag{4.12}$$

Next divide out $G = \gcd(E, H)$. Let

$$g_X = \deg_X G, \qquad g_Y = \deg_Y G.$$

Replace E, H by $E/G, H/G$, and replace the four degree bounds by

$$a_X - g_X, \quad b_X - g_X, \qquad a_Y - g_Y, \quad b_Y - g_Y.$$

The differences d_X, d_Y are unchanged. At most g_X points $a \in A$ satisfy $G(a, Y) \equiv 0$, and at most g_Y points $b \in B$ satisfy $G(X, b) \equiv 0$. Delete these points from the two evaluation sets. On every remaining slice, cancellation of G preserves (4.9) and (4.10). Moreover,

$$\frac{b_X - g_X}{|A| - g_X} \le \frac{b_X}{|A|}, \qquad \frac{b_Y - g_Y}{|B| - g_Y} \le \frac{b_Y}{|B|},$$

because (4.11) implies $b_X < |A|$ and $b_Y < |B|$. Thus all hypotheses remain valid. Condition (4.12) also remains valid after subtracting the gcd degrees. Renaming the reduced objects, we may assume that E, H are coprime and that (4.12) holds.

Regard E, H as polynomials in X over $\mathbb{F}(Y)$. Consider

$$\Phi : \mathbb{F}(Y)[X]_{<a_X} \oplus \mathbb{F}(Y)[X]_{<b_X} \longrightarrow \mathbb{F}(Y)[X]_{<a_X+b_X},$$

$$\Phi(R, S) = RH + SE. \tag{4.13}$$

The source and target have the same dimension. The map is injective. Indeed, if $RH + SE = 0$, coprimality gives $E \mid R$ and $H \mid S$, so

$$R = ET, \qquad S = -HT$$

for some $T \in \mathbb{F}(Y)[X]$. If $\deg_X E = a_X$, the inequality $\deg_X R < a_X$ forces $T = 0$; if instead $\deg_X H = b_X$, the inequality $\deg_X S < b_X$ forces $T = 0$. Thus $R = S = 0$.

Let $D_X(Y)$ be the determinant of the coefficient matrix of Φ in the monomial bases. It is nonzero. There are a_X columns coming from multiples of H, whose coefficients have Y-degree at most b_Y, and b_X columns coming from multiples of E, whose coefficients have Y-degree at most a_Y. Hence

$$\deg D_X \le a_X b_Y + a_Y b_X. \tag{4.14}$$

Fix $b \in B$. By (4.9), the image of the specialized map is contained in

$$E(X, b) \cdot \mathbb{F}[X]_{<b_X},$$

because $\deg R < a_X$ and $\deg r_b \le d_X = b_X - a_X$. This space has dimension at most b_X. The matrix has size $a_X + b_X$, so its rank deficiency is at least a_X. By Theorem 4.4,

$$(Y - b)^{a_X} \mid D_X(Y).$$

The points of B are distinct, and therefore

$$a_X |B| \le a_X b_Y + a_Y b_X. \tag{4.15}$$

Interchanging X and Y gives

$$a_Y |A| \le a_X b_Y + a_Y b_X. \tag{4.16}$$

Let $M = a_X b_Y + a_Y b_X$. From (4.15) and (4.16),

$$M = \frac{b_Y}{|B|}(a_X|B|) + \frac{b_X}{|A|}(a_Y|A|) \le \left(\frac{b_X}{|A|} + \frac{b_Y}{|B|}\right) M.$$

By (4.11), the coefficient is strictly less than one, so $M = 0$. Equations (4.15) and (4.16) force $a_X = a_Y = 0$. Hence the coprime reduction of E is constant, which means that the original E divides the original H. Finally, when $a_X = a_Y = 0$, the quotient degree bounds are $b_X = d_X$ and $b_Y = d_Y$; the reductions above did not change d_X, d_Y or the quotient. □

The quotient-degree bounds in (4.9) and (4.10) are essential: they force the specialized coefficient matrix to lose the required rank. The use of degree bounds rather than merely the actual degrees of E and H is the subtle point in the corrected form of the Polishchuk–Spielman argument [3].

4.4 Proof of the gluing and robustness theorems

Proof of Theorem 4.2. The vector space of polynomials of bidegree at most (e_A, e_B) has dimension $(e_A + 1)(e_B + 1) > s$. Hence there is a nonzero polynomial $E(X, Y)$ of these degree bounds that vanishes on all points of D.

Define a table

$$G(a, b) = E(a, b)r_b(a).$$

Outside D, the two local values agree; on D, the factor E vanishes. Thus

$$G(a, b) = E(a, b)c_a(b)$$

at every point of the grid.

Every first-coordinate slice of G has degree at most $e_A + d_A$, and every second-coordinate slice has degree at most $e_B + d_B$. Condition (4.6) implies that these degrees are smaller than the respective evaluation-set sizes. By Theorem 4.3, there is a polynomial $H(X, Y)$ with

$$\deg_X H \le e_A + d_A, \qquad \deg_Y H \le e_B + d_B,$$

and

$$H(X, b) = E(X, b)r_b(X), \qquad H(a, Y) = E(a, Y)c_a(Y).$$

Apply Theorem 4.5 with

$$a_X = e_A, \quad a_Y = e_B, \quad b_X = e_A + d_A, \quad b_Y = e_B + d_B.$$

Condition (4.6) is exactly (4.11), and the slice quotients have degrees at most d_A, d_B. Hence $H = EQ$ for a polynomial Q of bidegree at most (d_A, d_B).

If $r_b \neq Q(\cdot, b)$, then

$$E(X, b)\big(r_b(X) - Q(X, b)\big) = 0$$

as a polynomial in X. Hence $E(X, b) \equiv 0$, so $Y - b$ divides E. This can happen for at most e_B values of b. Similarly, there are at most e_A values of a for which $c_a \neq Q(a, \cdot)$.

Outside the original disagreement set D, the two local values agree. If this common value differs from $Q(a, b)$, then both a and b are exceptional. There are at most $e_A e_B$ such additional points. □

Proof of Theorem 4.1. Use the nearest slice polynomials chosen before (4.3), and put

$$\eta = \delta_A(F) + \delta_B(F).$$

By (4.4), their disagreement set has size

$$s \le \eta |A||B|. \tag{4.14}$$

If $\eta \ge 1/1024$, the theorem follows from the trivial estimate $\Delta(F, C_A \otimes C_B) \le 1$. Suppose $0 < \eta < 1/1024$; the case $\eta = 0$ is immediate.

Set

$$e_A = e_B = \lceil \sqrt{s} \rceil.$$

Then (4.5) holds. Put $M = |A|$ and $N = |B|$. The size assumption in (4.1) gives

$$\frac{1}{M} + \frac{1}{N} \le \frac{4}{\sqrt{MN}}.$$

Since $s \ge 1$, equation (4.14) implies $1/\sqrt{MN} \le \sqrt{\eta}$. Therefore

$$\frac{e_A}{M} + \frac{e_B}{N} \le (\sqrt{s} + 1)\left(\frac{1}{M} + \frac{1}{N}\right) \le 8\sqrt{\eta} < \frac{1}{4}.$$

Together with $d_A/M < 1/4$ and $d_B/N < 1/8$, this proves (4.6).

By Theorem 4.2, there is a codeword $Q \in C_A \otimes C_B$ such that the chosen local polynomials disagree with Q on at most $s + e_A e_B$ points. Since $s \ge 1$,

$$e_A e_B = \lceil \sqrt{s} \rceil^2 \le 4s.$$

Thus this exceptional set has relative size at most 5η. Comparing F first with its nearest first-coordinate slices and then with Q,

$$\Delta(F, Q) \le \delta_A(F) + 5\eta \le 6\eta.$$

This is stronger than (4.2) in the small-η case. Combining the two cases proves the theorem with $C_0 = 1024$. □

5 Soundness of the recursive verifier

Let $D(k)$ be the smallest number such that, for every dimension-k evaluation space, every word f, and every purported proof,

$$\Delta(f, \mathrm{RS}_{1/8}) \le D(k) \Pr[\text{reject}]. \tag{5.1}$$

For $k \le k_0$, the verifier checks the word exactly, so $D(k) = 1$ is enough.

Fix a recursive call of dimension $k > k_0$, with the notation of Section 3. Let

$$\varepsilon_A = \mathbb{E}_{v \in V}[\text{rejection probability of the recursive test on } R_v],$$

$$\varepsilon_B = \mathbb{E}_{x \in A}[\text{rejection probability of the recursive test on } B].$$

The parent rejection probability is

$$\varepsilon = \frac{1}{2}(\varepsilon_A + \varepsilon_B). \tag{5.2}$$

For each $v \in V$, the induction hypothesis gives a polynomial $r_v(T)$ of degree less than m such that

$$\rho_v := \Delta_{R_v}(\widehat{h}(\cdot, q_U(v)), r_v) \le D(r+3)\varepsilon_v,$$

where ε_v is that child's rejection probability. Hence

$$\overline{\rho} := \mathbb{E}_v \rho_v \le D(r+3)\varepsilon_A. \tag{5.3}$$

Similarly, for each $x \in A$, there is a polynomial $c_x(Y)$ of degree less than $n/8$ with

$$\kappa_x := \Delta_B(\widehat{h}(x, \cdot), c_x) \le D(s)\varepsilon_x,$$

and therefore

$$\overline{\kappa} := \mathbb{E}_x \kappa_x \le D(s)\varepsilon_B. \tag{5.4}$$

Restrict $\widehat{h}$ to $A \times B$, and call the resulting table g. Since $|R_v| = 8m$ and $|A| = 4m$, restricting from R_v to A can at most double the relative error. Thus

$$\delta_A(g) \le 2\overline{\rho}, \qquad \delta_B(g) \le \overline{\kappa}.$$

The set A has size $4m$, while $|B| = n$ and $n/m \in \{1, 2\}$. Hence $|A|/|B| \in \{2, 4\}$, and the hypotheses of Theorem 4.1 hold. We obtain a polynomial $Q(X, Y)$ with

$$\deg_X Q < m, \qquad \deg_Y Q < n/8, \tag{5.5}$$

such that

$$\tau := \Delta_{A \times B}(g, Q) \le C_0(2\overline{\rho} + \overline{\kappa}). \tag{5.6}$$

It remains to recover the original word on L. Call $v \in V$ bad if

$$r_v(X) \ne Q(X, q_U(v))$$

as polynomials. Both sides have degree less than m. If they are distinct, they agree at at most $m - 1$ of the $4m$ points of A, and therefore

$$\Delta_A(r_v, Q(\cdot, q_U(v))) > \frac{3}{4}.$$

The triangle inequality and averaging over v give

$$\Pr_v[v \text{ is bad}] \le \frac{4}{3}(2\overline{\rho} + \tau). \tag{5.7}$$

For a good v, the two polynomials are identical. The input fiber $U + v$ has m points, whereas R_v has $8m$ points, so the relative distance between f and $Q(\cdot, q_U(v))$ on a good input fiber is at most $8\rho_v$. Charging every point of a bad fiber as an error,

$$\Delta_L(f,\, x \mapsto Q(x, q_U(x))) \le 8\overline{\rho} + \frac{4}{3}(2\overline{\rho} + \tau). \tag{5.8}$$

By (5.6),

$$\Delta_L\big(f,\, x \mapsto Q(x, q_U(x))\big) \le K(\overline{\rho} + \overline{\kappa}) \tag{5.9}$$

for an absolute constant K.

Finally, define

$$P_Q(T) = Q(T, q_U(T)).$$

Since $\deg q_U = m$, equation (5.5) gives

$$\deg P_Q \le (m-1) + m\left(\frac{n}{8} - 1\right) = \frac{mn}{8} - 1. \tag{5.10}$$

Thus $P_Q|_L \in \mathrm{RS}_{1/8}(L)$. Combining (5.3), (5.4), and (5.9),

$$\Delta_L(f, \mathrm{RS}_{1/8}(L)) \le K\big(D(r+3)\varepsilon_A + D(s)\varepsilon_B\big).$$

Using (5.2),

$$D(k) \le 2K \max\{D(r+3), D(s)\}. \tag{5.11}$$

Both child dimensions are at most $k/2 + 3$. The recursion has depth $O(\log k)$, so (5.11) implies

$$D(k) \le (k+1)^C$$

for some absolute constant C. This proves soundness.

6 Proof length, randomness, and queries

Let $\Pi(k)$ be the number of field symbols in the auxiliary proof for a dimension-k input. The function h has exactly $7N$ symbols: the n sets $R_v \times \{q_U(v)\}$ contain $8N$ positions in total, and the N positions of Γ are supplied by the input rather than repeated in the proof. There are $n = 2^s$ recursive proofs on spaces of dimension $r+3$, and $|A| = 4m = 2^{r+2}$ recursive proofs on spaces of dimension s. Hence

$$\Pi(k) \le 7 \cdot 2^k + 2^s \Pi(r+3) + 2^{r+2}\Pi(s). \tag{6.1}$$

Writing $T(k) = \Pi(k)/2^k$,

$$T(k) \le 7 + 8T(r+3) + 4T(s). \tag{6.2}$$

A routine induction gives $T(k) \le (k+1)^{C_1}$ for some absolute constant C_1. Therefore

$$\Pi(k) \le N(\log N)^{O(1)}.$$

Let $R(k)$ be the maximum number of random bits used from a dimension-k call onward. A first-direction call uses one branch bit, s bits to choose v, and then $R(r+3)$ bits. A second-direction call uses one branch bit, $r+2$ bits to choose a point of A, and then $R(s)$ bits. Thus

$$R(k) \le 1 + \max\{s + R(r+3),\ r + 2 + R(s)\}. \tag{6.3}$$

Since the recursive dimensions are roughly halved,

$$R(k) = k + O(\log k) = \log N + O(\log \log N).$$

The verifier follows only one recursive branch. It reads no oracle symbols until reaching the constant-dimensional base case, where it reads the entire current word. Hence it makes only a constant number of field queries. This completes the proof of Theorem 1.1.

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